PENGQUIN:

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PErsoNal Genome and QUery in healthcare and clinical practice

PhD Candidate: Elias Crum

VITO 18-month jury
Nov 20th, 2025

Advisors: Dr. Gökhan Ertaylan[1], Dr. Bart Buelens[1], Dr. Ruben Taelman[2], Prof. Dr. Ruben Verborgh[2]

1 = vito

2 = GHENT UNIVERSITY

Presentation Overview

Project Review

~24 Month Progress

Current Undertakings

Remaining PhD Plans

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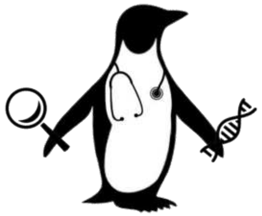
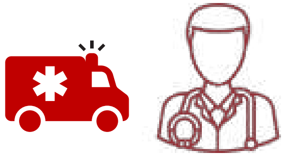
Genomic Data

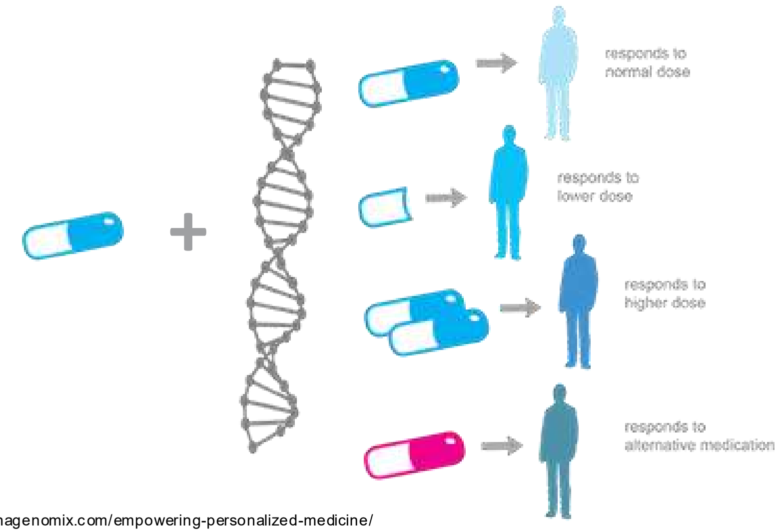




Clinical Care

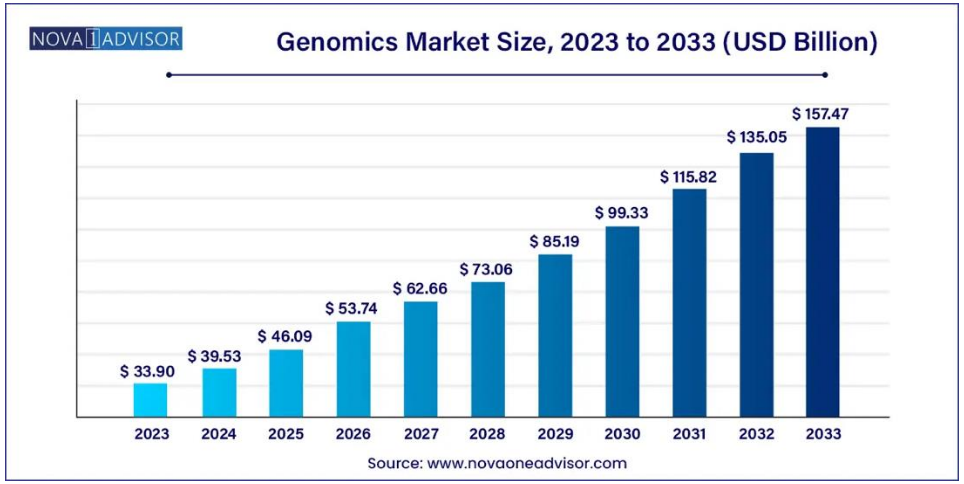
Genomics + Healthcare

A *Example: Pharmacogenomics*



https://alphagenomix.com/empowering-personalized-medicine/

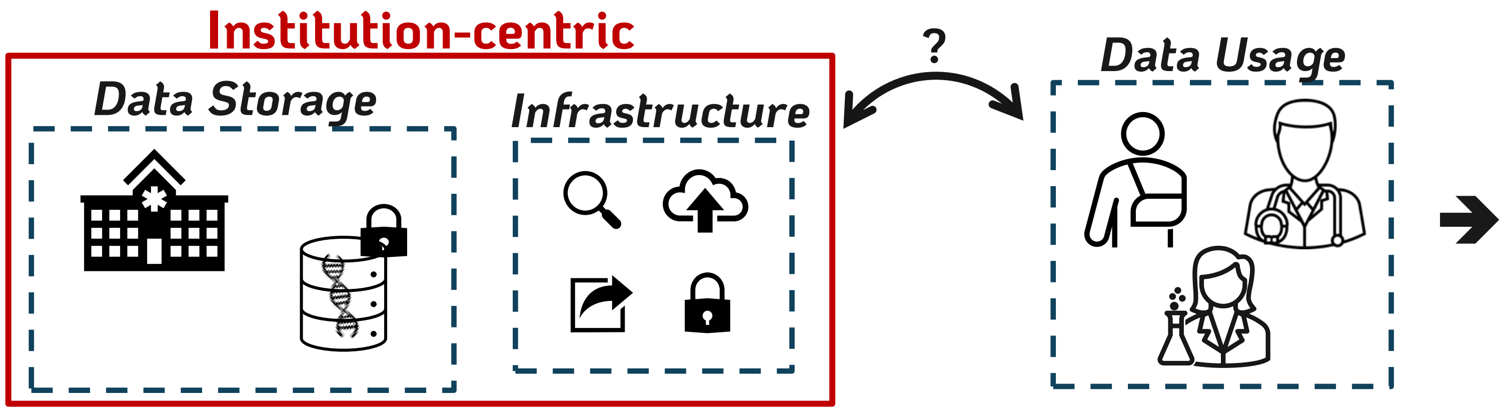
B *Genomics market is growing RAPIDLY*



https://www.novaoneadvisor.com/report/genomics-market

C

Problem:



Low Efficiency
+
High Costs

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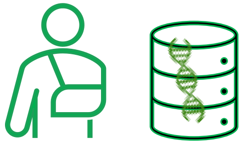
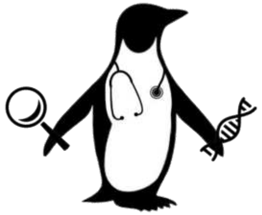
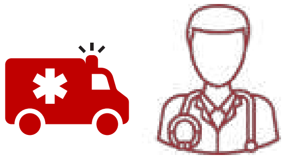
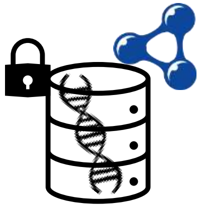


Genomic Data





Clinical Care



Private, Semantic Linked Data

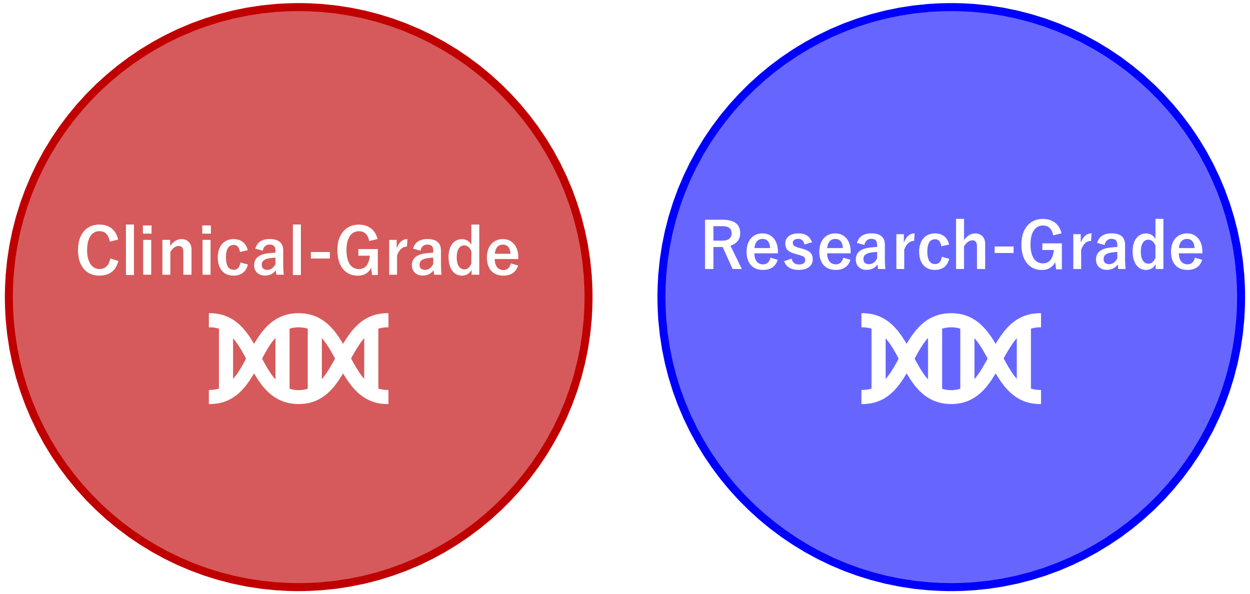
Query Capabilities

GENOME Pods

Framework + App

Genomic Data for Clinical Use

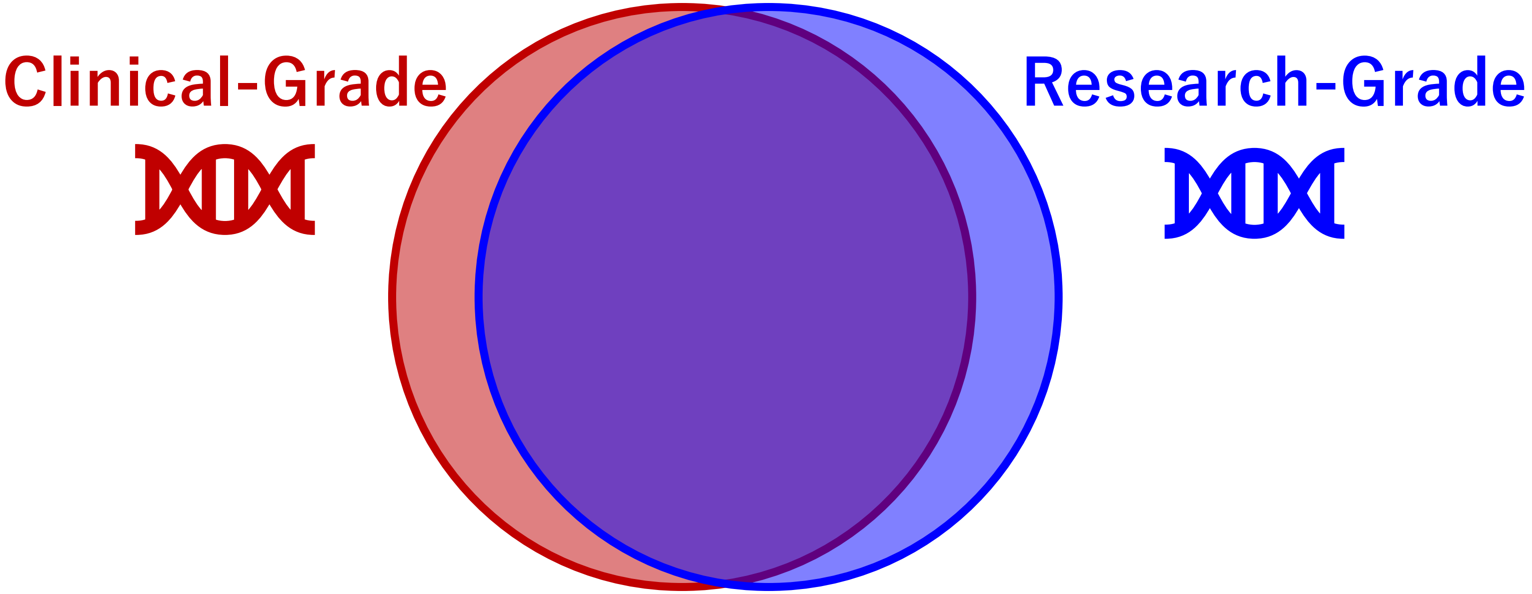
- Format is use-case specific
- Whole Genome Sequence (WGS) data
- Large Projects: 1+ Million Genomes (EU)[1]



[1] https://digital-strategy.ec.europa.eu/en/policies/1-million-genomes

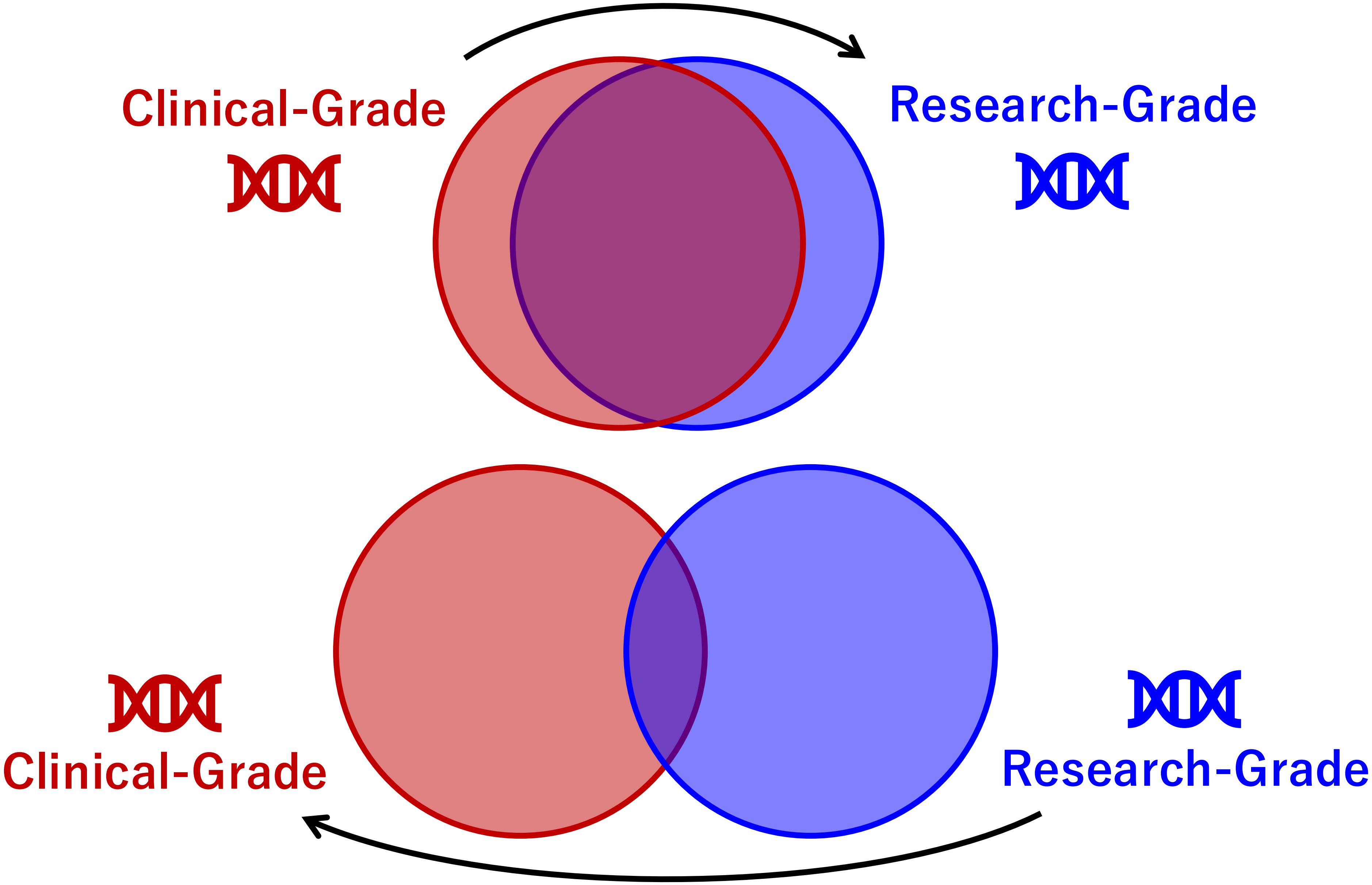
Genomic Data for Clinical Use (assumption)

- Format is use-case specific
- Whole Genome Sequence (WGS) data
- Large Projects: 1+ Million Genomes (EU) [1]



[1] https://digital-strategy.ec.europa.eu/en/policies/1-million-genomes

Genomic Data for Clinical Use (reality)



The current landscape

Intended use is VERY important

Clinical use ≠ Research use

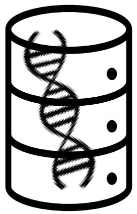
** Funding goes predominantly to RESEARCH-GRADE **

- Cheaper
- Fewer regulatory challenges
- Funders / Policy makers misunderstanding

** But how do we “translate” to clinical use?? **

Sharing Genomic Data – Technology Considerations

Key Implementation Factors



Data Storage

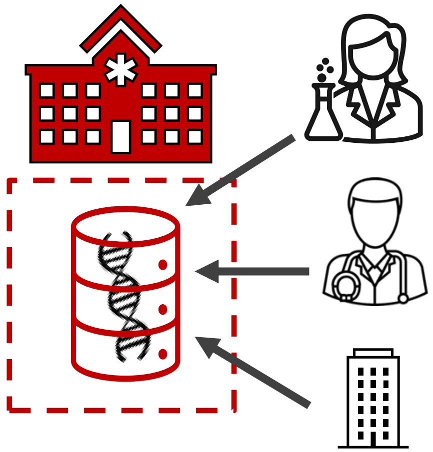
Considerations:

- patient consent
- data security
- access controls

&

Considerations:

- consent changes
- governance
- infrastructure



Data Sharing

A more in-depth analysis of these challenges + potential solutions detailed in an upcoming publication in *Computers in Biology and Medicine*

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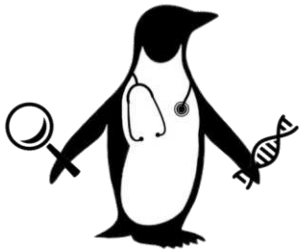

Genomic Data → Clinical Care

Framework + App

1. Genomic Data Interaction

Semantic Representation

2. Semantify Genomic Data

Query Capabilities

1. Data Interaction Framework

A Web Application to:

- *Upload Data to a Solid Pod*
- *Browse contents of a Solid Pod*
- *Edit Privacy of (and share) data in a Solid Pod*
- *Query Data in Solid Pods + SPARQL Endpoints*
  - ↳ *Includes a query cache for storing queries and results*

*** Enables patient, provider, and auditor control over data ***

What is a Solid Pod?

Secure, Decentralized Data Storage Specification

- *Pod exists on a Server (accessible to the Web)*
- *Can store files, RDF data, and Linked Data*
- *Provides control policies for stored data*

*** Enables secure, authenticated sharing via HTTP protocols ***

https://solidproject.org/about

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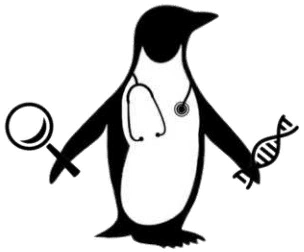


Genomic Data

Clinical Care

Framework + App

1. **Genomic Data Interaction**

Semantic Representation

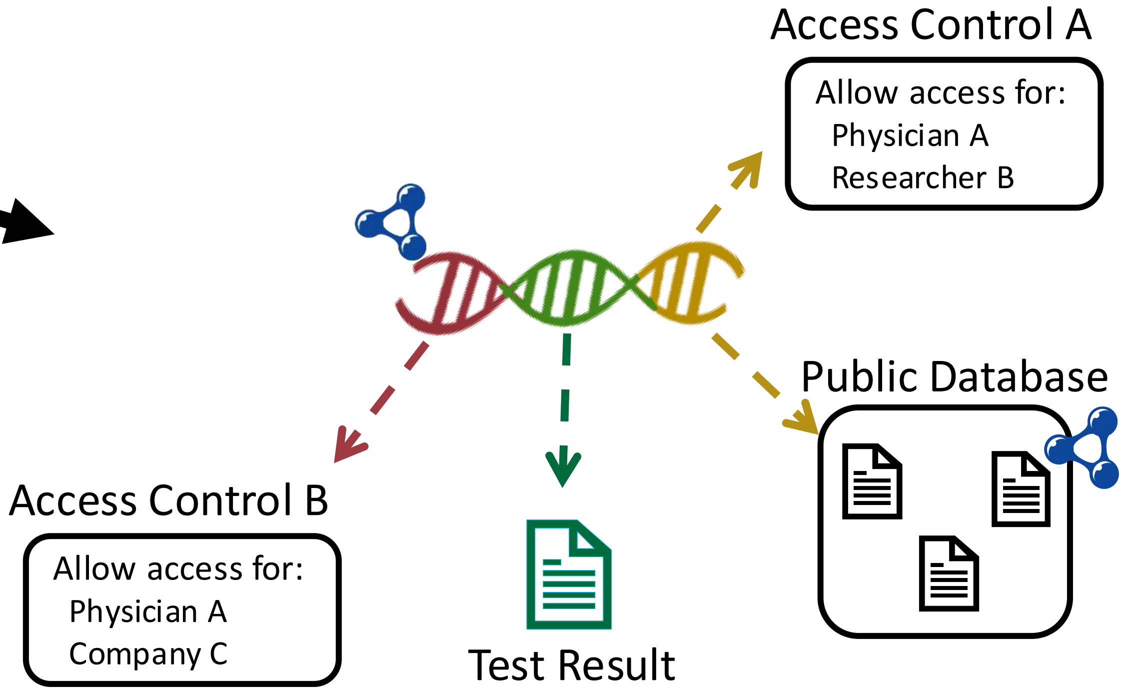
2. **Semantify Genomic Data**

Query Capabilities

2. Genomic Data Representation

From a file-based approach

To a granular, dynamic RDF approach



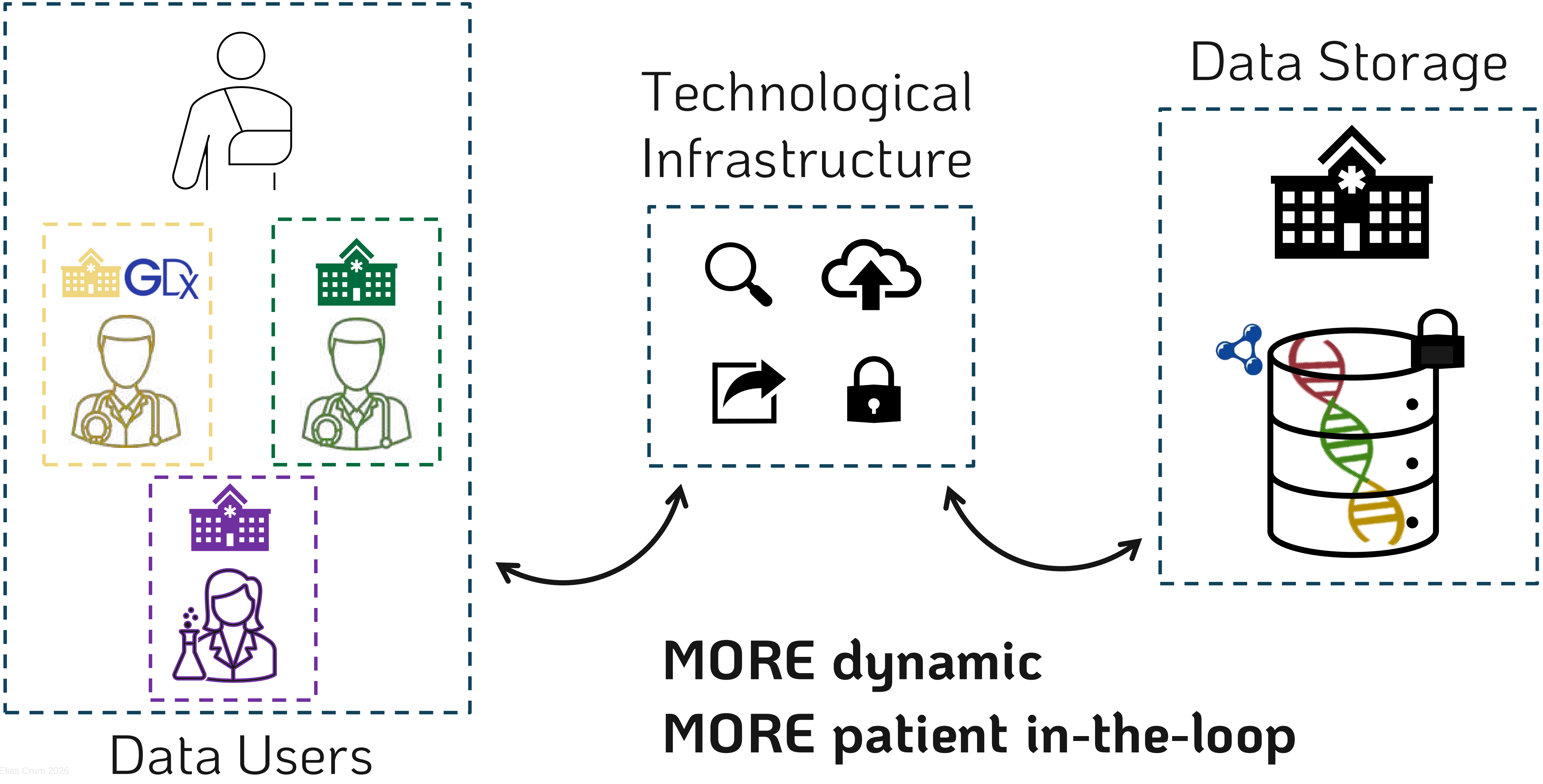
Conversion using existing ontologies

Policy definition via ODRL[1]

Implementation within Solid Pod Env

[1] https://www.w3.org/TR/odrl-model/

Future Possibilities?



MORE dynamic

MORE patient in-the-loop

Thank you!

More PhD specifics

LinkedIn

GitHub

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Informal Demo / Tutorial Later Today

(for those interested)

Solid Cockpit

knowledgeonwebscale.github.io/solid-cockpit/

Supplemental Slides

Genomic Data

Clinical-Grade

- Higher read quality
- More expensive
- Stricter use-regulations

Research-Grade

- Lower read quality
- Less expensive
- Flexible use-regulations

Dimension	Clinical Grade Genome Data Sharing	Research Grade Genome Data Sharing
1. Primary Purpose and Scope	Patient care and medical decision-making.	Discovery and hypothesis-driven studies.
2. Data Accuracy and Quality	Stringent quality control (QC) and validation.	High but no absolute QC standards.
3. Regulatory Requirements	Subject to medical regulatory frameworks.	Subject to research ethics regulations.
4. Consent Requirements	Clinical-informed consent.	Research-informed consent.
5. Privacy and Confidentiality	Highest level of patient privacy required.	Strong but variable privacy measures.
6. Security Controls	Robust, enterprise-grade security.	Secure but more flexible.
7. Data Governance and Oversight	Hospital/clinic-based governance.	Institution or consortium-based governance.
8. Ethical Considerations	Focus on direct patient benefit and “do no harm.”	Focus on scientific advancement and participant welfare.
9. Infrastructure Requirements	Highly reliable, clinically validated systems.	Flexible, scalable research infrastructure.
10. Architecture Requirements	Standardized, interoperable clinical systems.	Flexible, modular architectures.
11. Legal Liabilities	High.	Moderate.
12. Interoperability	High necessity for clinical integration.	Variable.
13. Data Ownership and Control	Patient-oriented ownership with institutional custodianship.	Institution or sponsor-owned.
14. Data Sharing Scope	Limited to necessary clinical stakeholders.	Broad or consortium-based.
15. Cost and Funding Models	Healthcare reimbursement or institutional budgets.	Grant-funded or institutional research budgets.

Genomic Sequence Data Sharing for Clinical Practice: A Scoping Review

Aim: Identify existing implementations, guidelines, and challenges of clinical sharing of genomic data.

Methods: Reviewed 55 studies, 57 national genomics projects, and 20 major genomics companies.

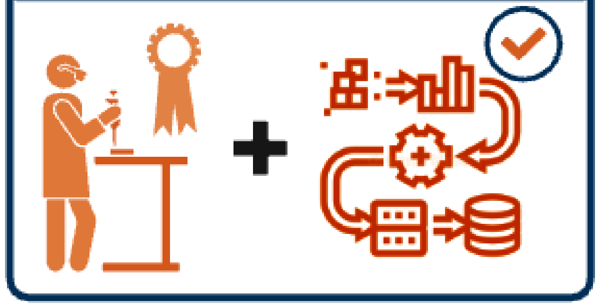
Results: Clinical genome sharing implementations exist, yet scalability and widespread adoption remain limited.

Suggested steps forward:

Harmonize legal bases for secondary clinical use of research data

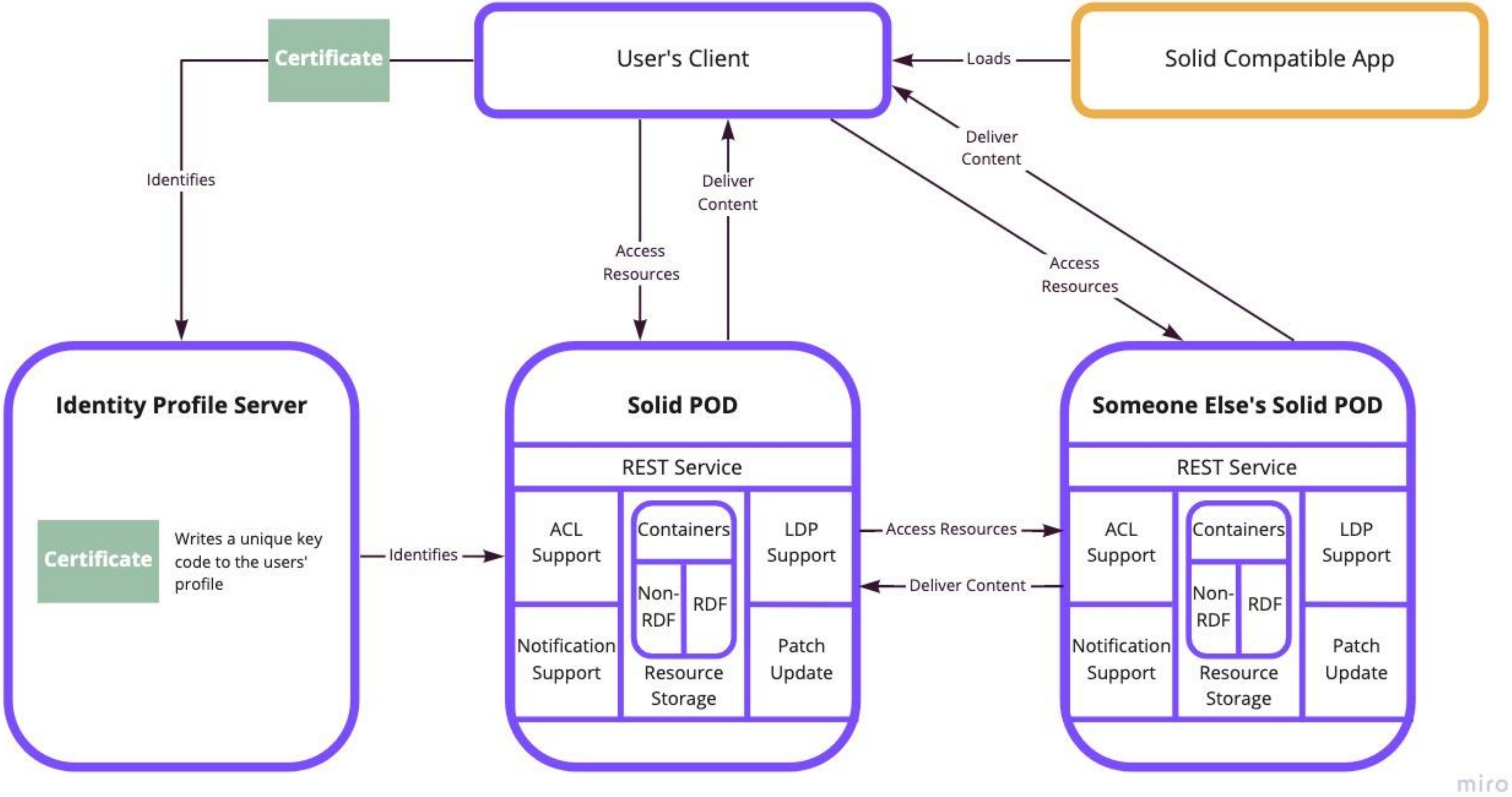
Mandate (GA4GH) technical standards

Scale laboratory accreditation & continual pipeline validation



Align economic incentives

Solid's Architecture



Solid Pod Implementations?

BBC MyPDS App

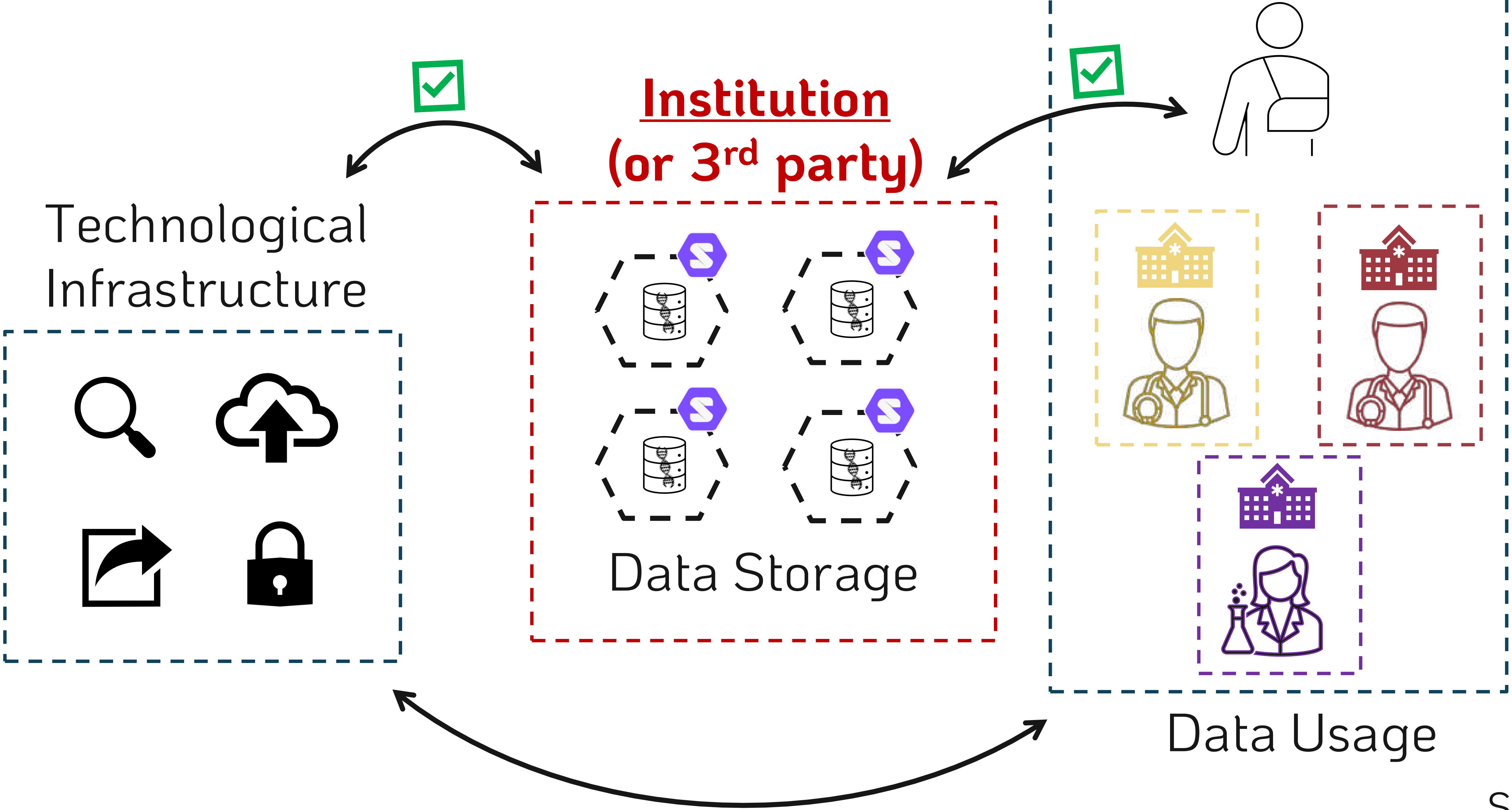
Flemish Data Utility Company

health data vaults

Flemish Smart Dataspace

Solid Lab Use-cases

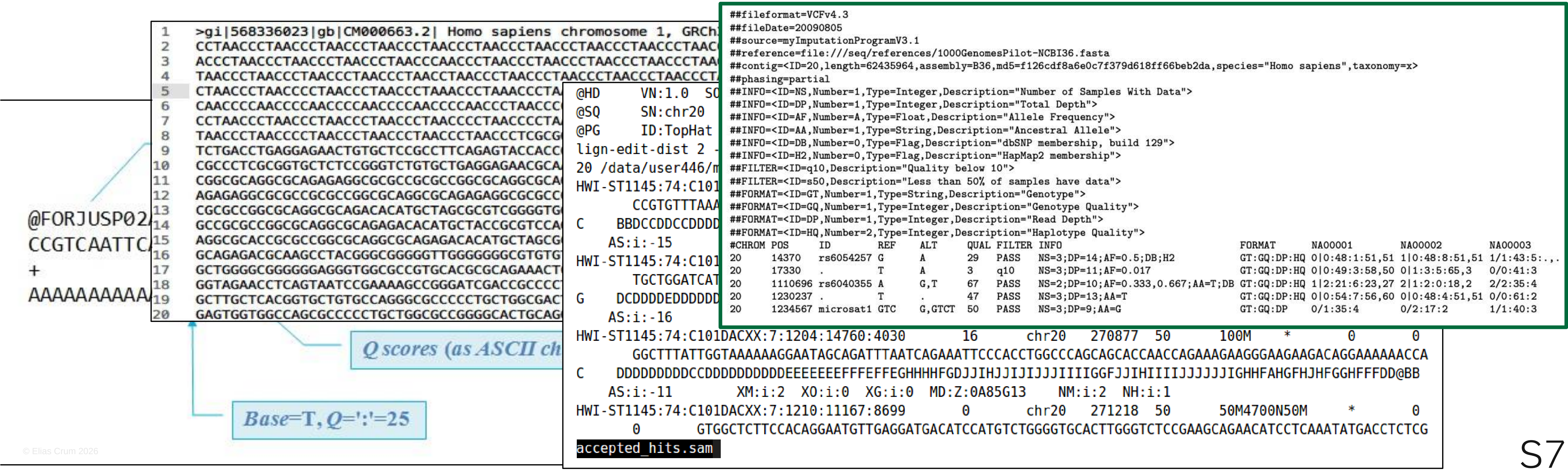
A Production Solid Solution



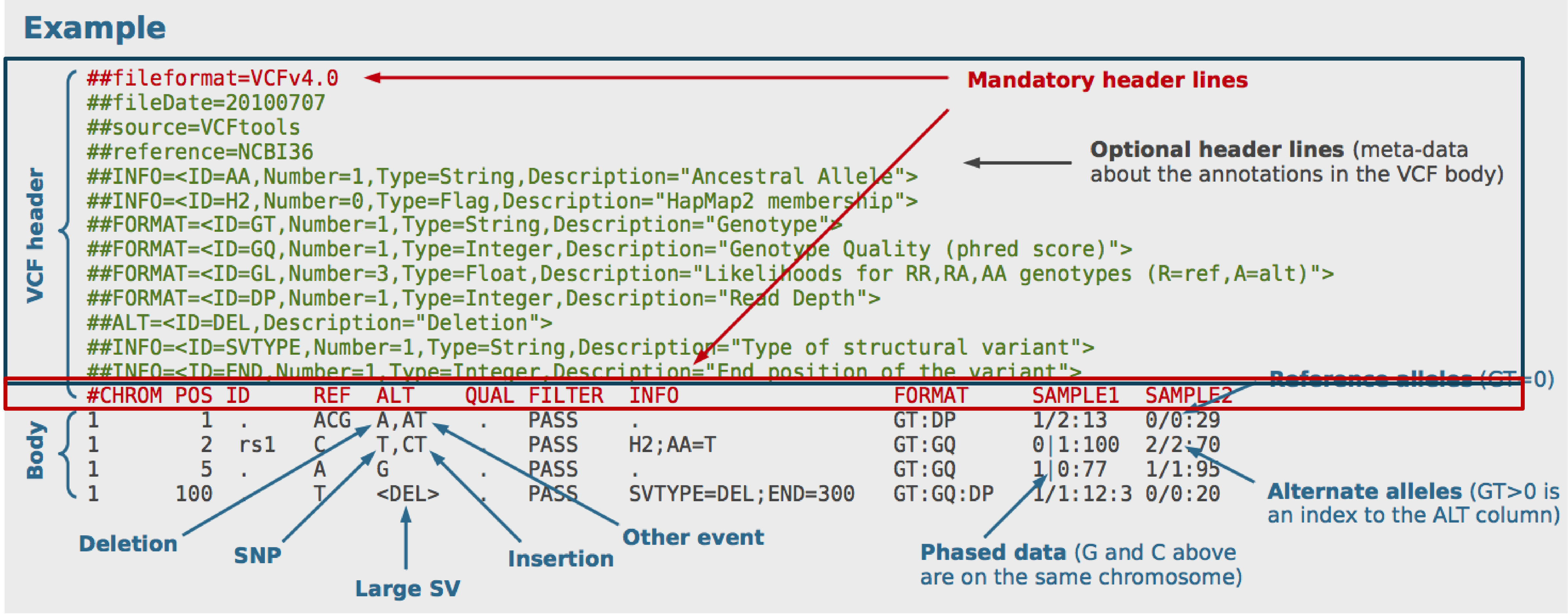
Semantic VCF – *Sequencing Data Formats Intro*

FASTQ ✚ **FASTA** → **SAM/BAM** → **VCF**

FASTQ	FASTA	SAM/BAM	VCF
Raw Sequencing Outputs	Contiguous DNA sequence	FASTQ reads aligned to a FASTA reference	Areas in SAM where sample and reference differ



What does a VCF File Look like?

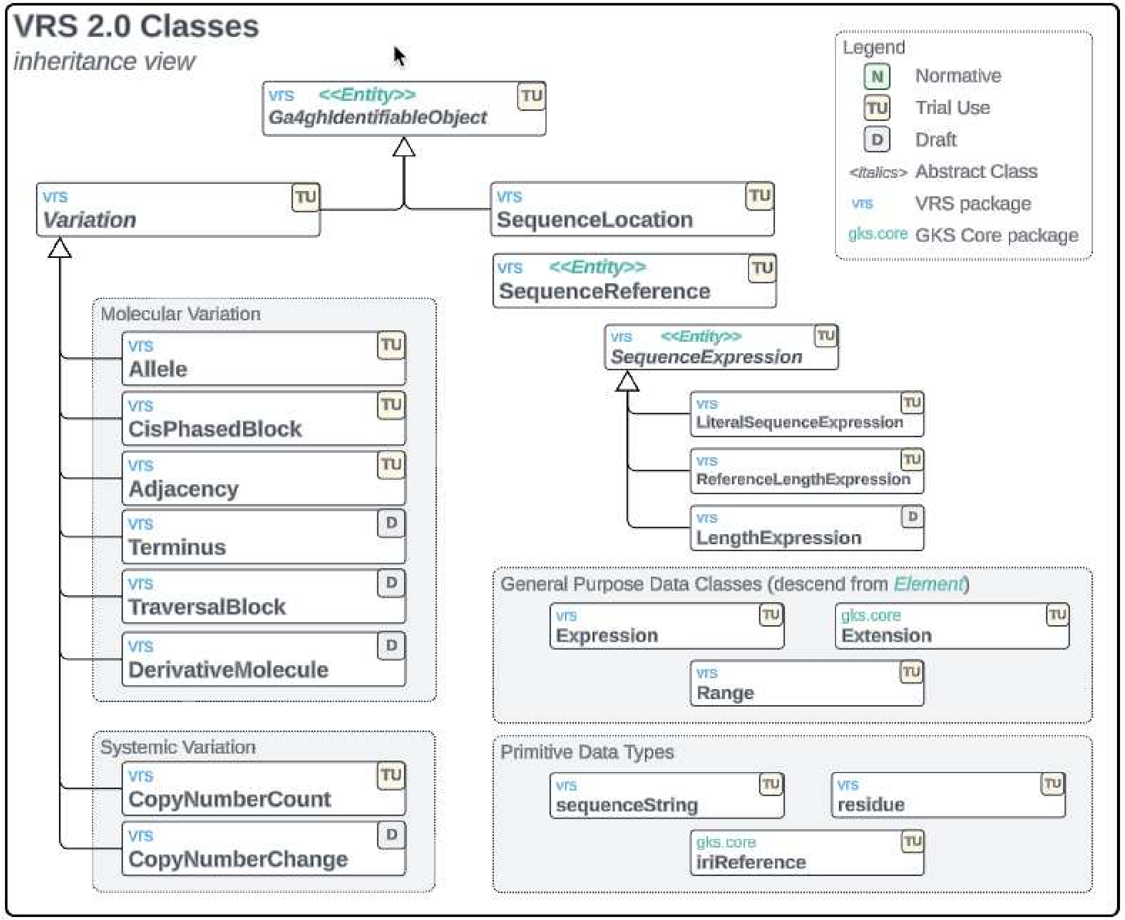


https://davetang.github.io/learning_vcf_file/

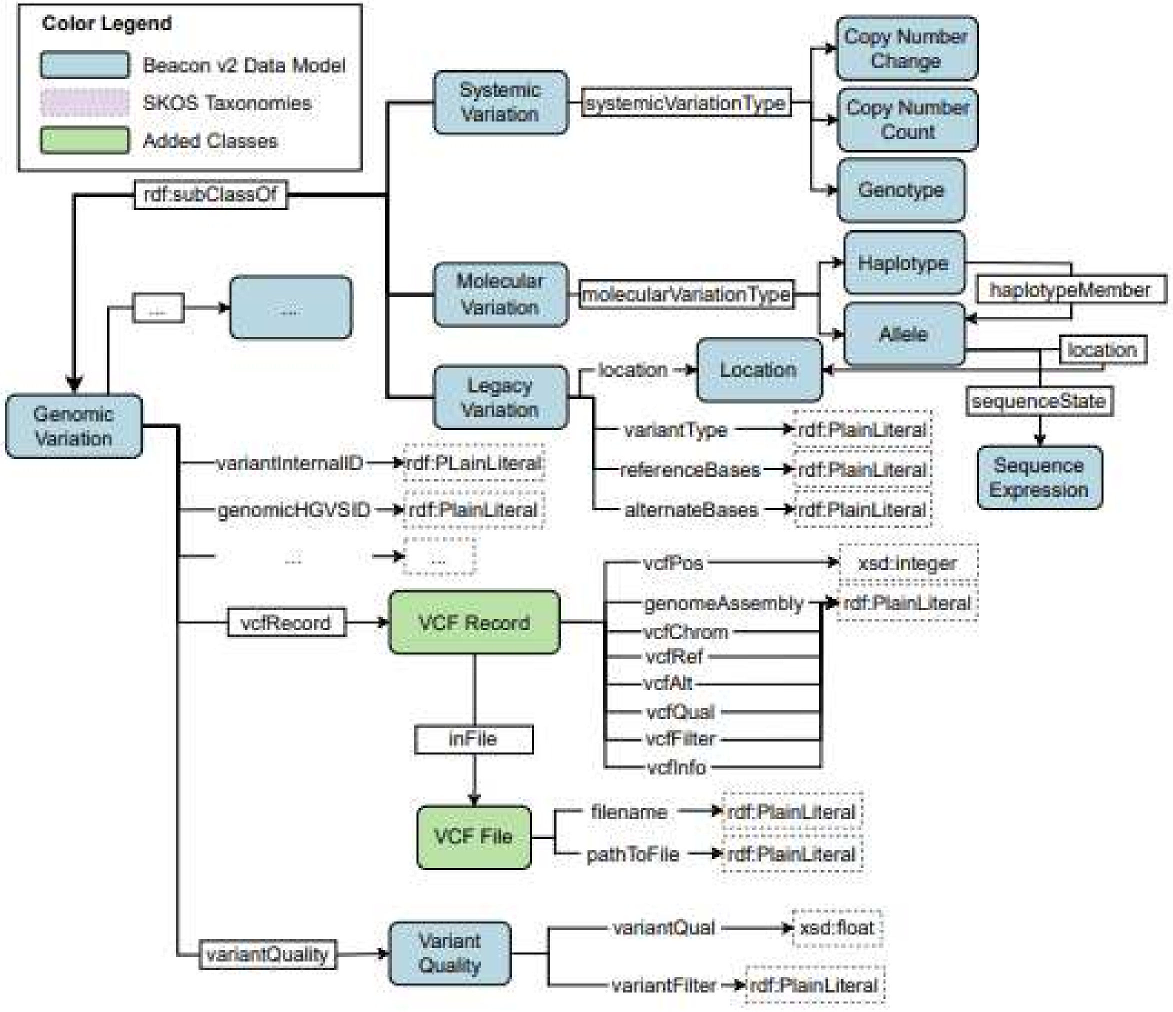
VCF Spec: https://samtools.github.io/hts-specs/VCFv4.2.pdf

Representing VCF Semantically?

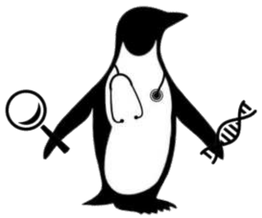
GA4GH Variant Schema:



HERO Genomics Ontology:



Envisioned Product Evolution



PhD End | Short Term | Medium Term | Long Term

PhD Deliverable

Start-up / Product development
+
Hospital implementation

Integrate with EHR
+
Increase application domains

Increase 3rd party application support
+
Scale to Europe and beyond